

Architectural design of our WarZone game

Our design is split into two modules, Models and Controllers. The models consist of the following classes: Player, Orders, Country, Continent, WarMap.

The Orders class stores the information about the orders that players make.

The Player class stores its orders, the number of reinforcements it can currently place as well as its owned countries.

The Country class simply stores an ID, name, continent ID, army count, and list of neighboring countries.

The Continent class stores an ID and army bonus.

Lastly the WarMap stores all of the countries and continents as well as an adjacency list of the entire WarMap, it also holds important functions for validating and saving maps.

The Controllers are made up of the following classes: MapEditor, MainGameLoop and GameEngine.

The MapEditor is responsible for the controlling flow when a map is being edited from the console, it holds both the read and edit map functions to load from a file, this allows starting from scratch with a new file, or reading a pre-existing file and editing on top of it.

The GameEngine is responsible for game set-up or passing control to the map editor, it requires a user to load a valid map and add an adequate number of players, once these conditions are met it can use the assign countries function to start the game and pass control to the MainGameLoop.

The MainGameLoop class is responsible for calculating the number of reinforcements, and having the deploy orders taken from the user and then executing these orders.